SSR

PRESENTATION

ARP SPOOFING



CONTENT:

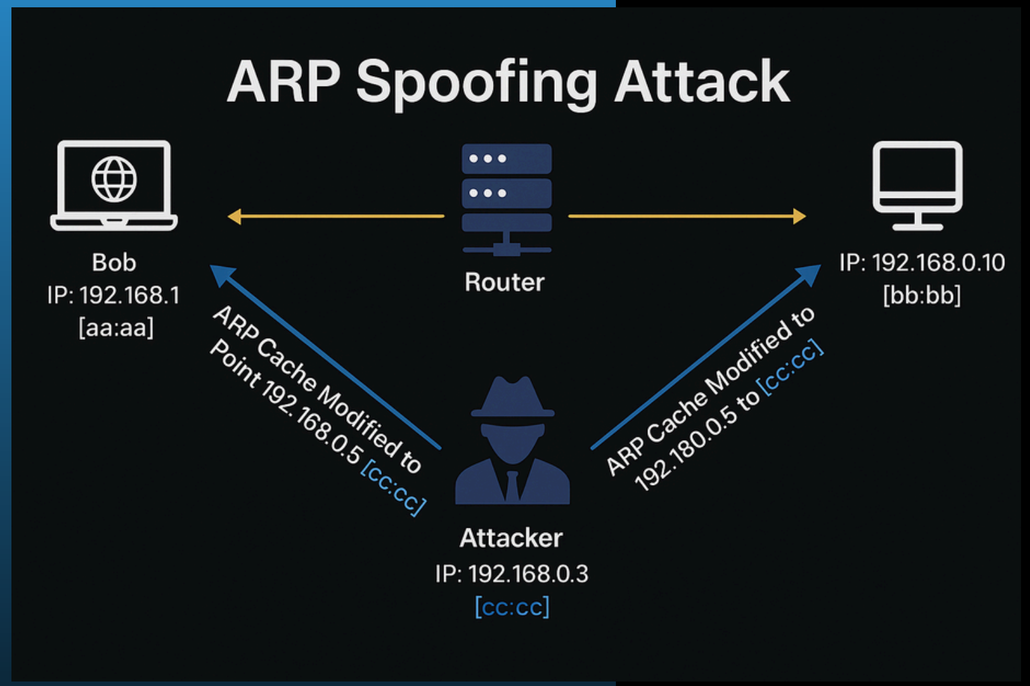
What is ARP Spoofing?

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WHAT IS ARP SPOOFING?

- ARP Spoofing (also known as ARP Poisoning) is a cyberattack where an attacker sends falsified Address Resolution Protocol (ARP) messages over a local network. This manipulates the network's ARP tables, allowing the attacker to intercept, modify, or stop data traffic between devices.

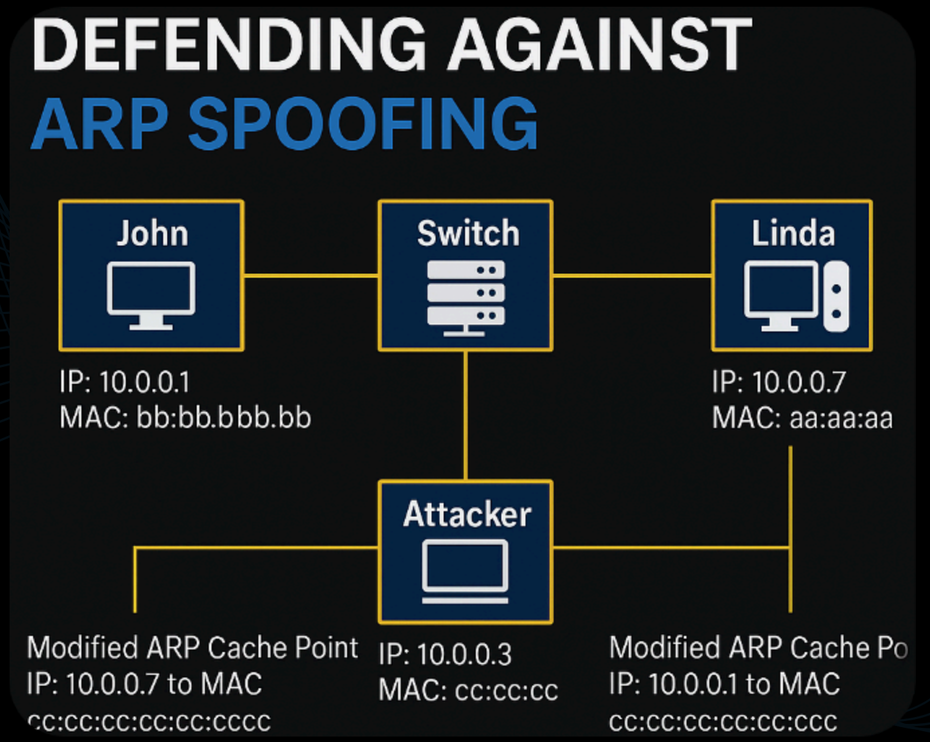
HOW ARP SPOOFING WORKS

ARP Basics:

- ARP isusedtomap IP addresses to MAC addresses on a local network.
- Devices maintain an ARP cache (table) to store these mappings.

The Attack Process :

- The attacker sends fake ARP replies to the victim's devices, claiming:
  - "I am the router" (to the victim).
  - "I am the victim" (to the router).
- This tricks both the victim and the router into sending traffic through the attacker's machine.



Man-in-the-Middle (MitM) :

- Oncethe ARPcacheispoisoned,the attacker can:
- Eavesdrop on traffic (passive attack).
- Modify data (active attack).
- Block communication (Denial-of-Service).

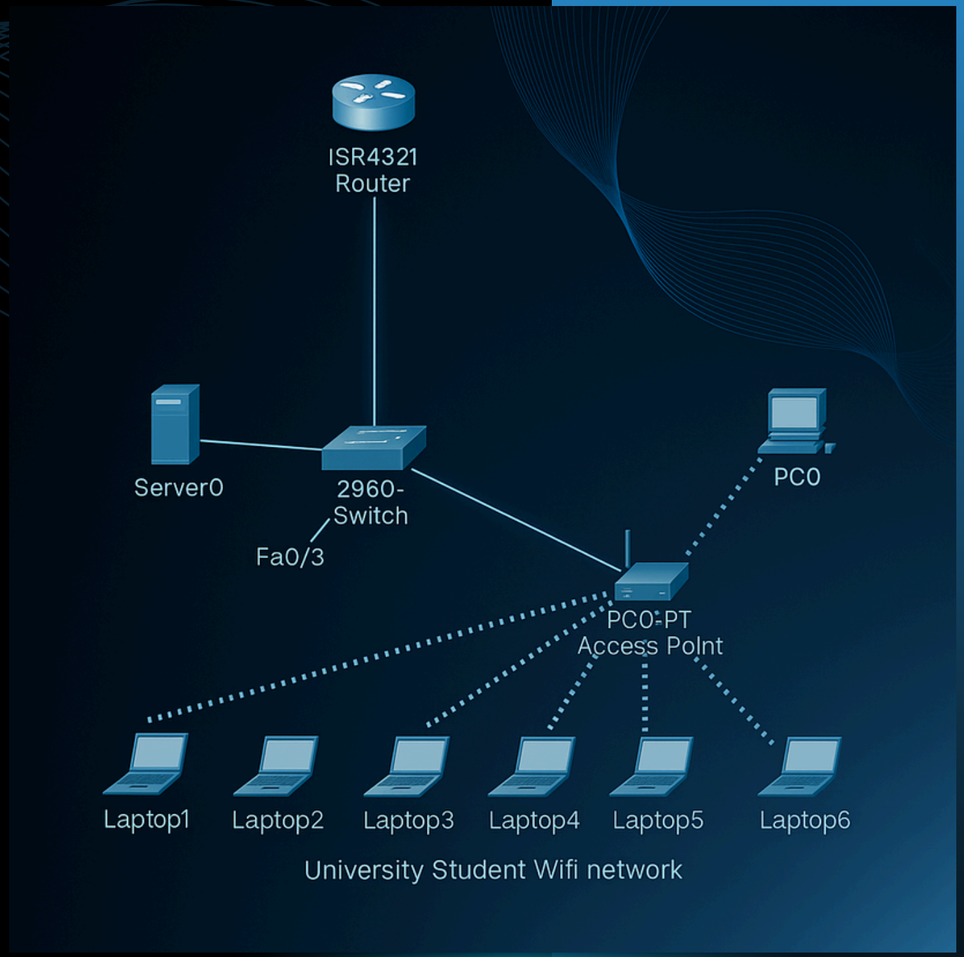
Purposes of ARP Spoofing :

- Session Hijacking Stealing login sessions (cookies for login to target personel account).
- Data Interception Capturing passwords, emails.
- Denial-of-Service(DoS) Disrupting network communication.
- Attacks can Modifying traffic in transit.



INFASTRUTURE TO TARGET NETWORK

- Network Environment:
- The university offers open-access Wi-Fi for all students and staff.
- simple password shared with all students.
- All connected devices share the same Layer 2 (data link) network.
- Devices can see and communicate with each other directly.
- Vulnerabilities:
- No client isolation: Any connected user can scan and interact with others.
- Dynamic ARP handling: Devices trust any ARP response received.
- No centralized ARP protection like DHCP snooping or dynamic ARP inspection.
- Many students are unaware of network threats, increasing the chance of a successful attack.





HOW TO ATTACK

DEMONSTRATION

Tools we use:

- **KALI LINUX**
- **BETTERCAP**
- **ETTERCAP**

WHAT IS KALI LINUX

- KaliLinux isapenetration testing and ethical hacking operating systembasedon Debian.
- It comes pre-installed with hundreds of security tools for tasks like network scanning, password cracking, wireless attacks, and more.
- Popular among cybersecurity professionals and researchers.





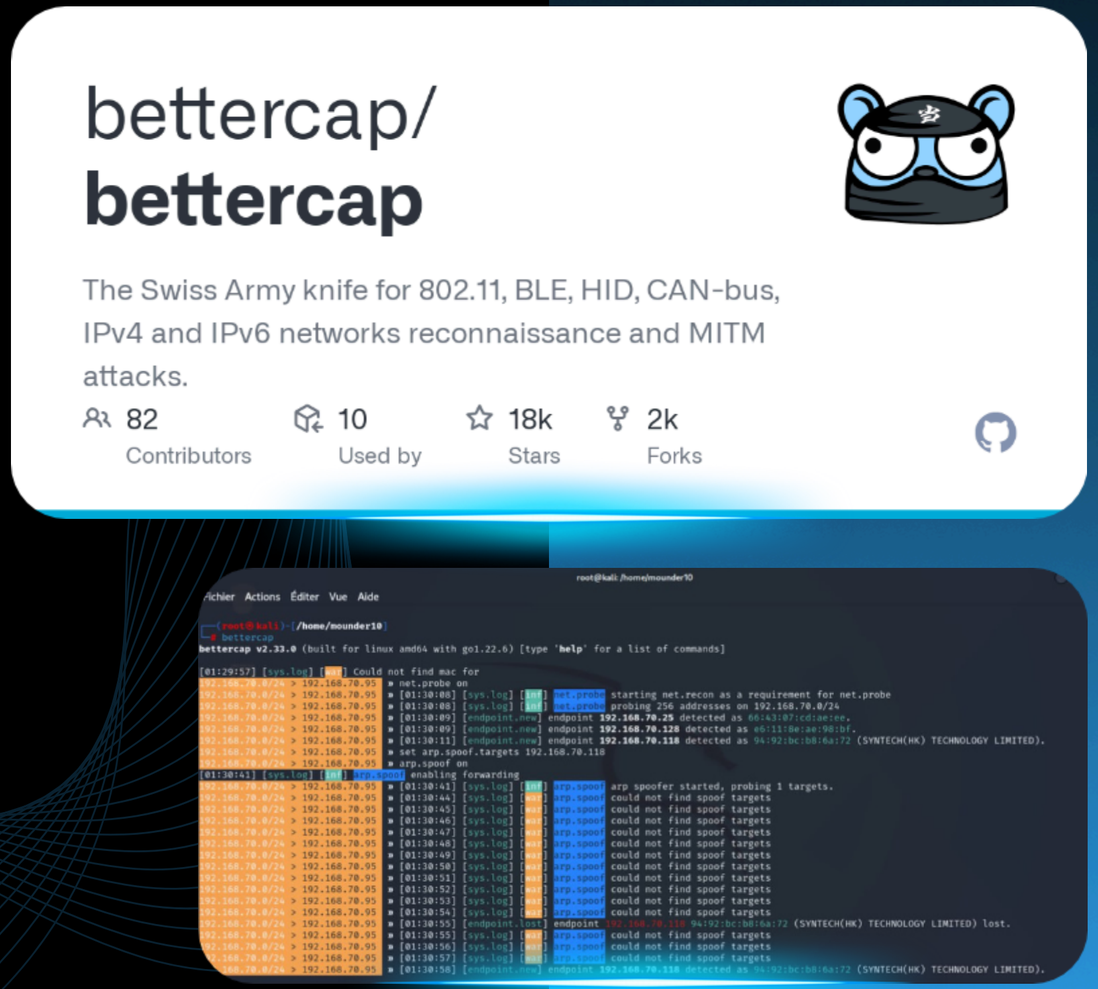
WHAT IS ETTERCAP



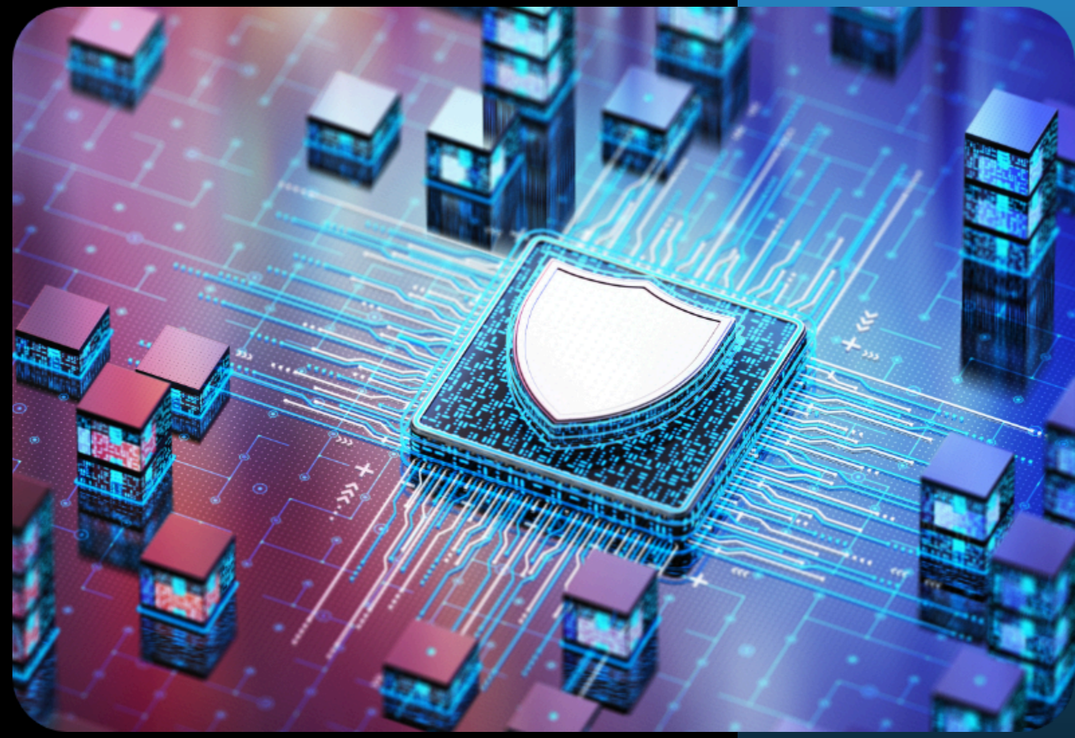
- Ettercapis anetworksecuritytool for man-in-the-middle (MitM) attacks on LAN.
- It supports ARP spoofing, sniffing passwords, and active traffic manipulation.
- Has both GUI and CLI interfaces.
- an filter or inject data into live connections, ideal for educational demonstrations

WHAT IS BETTERCAP

- Bettercap is a modern, more advanced alternative to Ettercap
- Built for real-time packet manipulation, sniffing, and injection in networks
- Includes modules for Wi-Fi hacking, HTTPS stripping, credential harvesting, and more



DEFENDING AGAINST ARP SPOOFING



HowtoDetect ARPspoofing:

- ARP Monitoring Tools (e.g., Wireshark, ARPwatch).
- Unusual Network Activity (e.g., slow connections, unexpected MAC addresses).

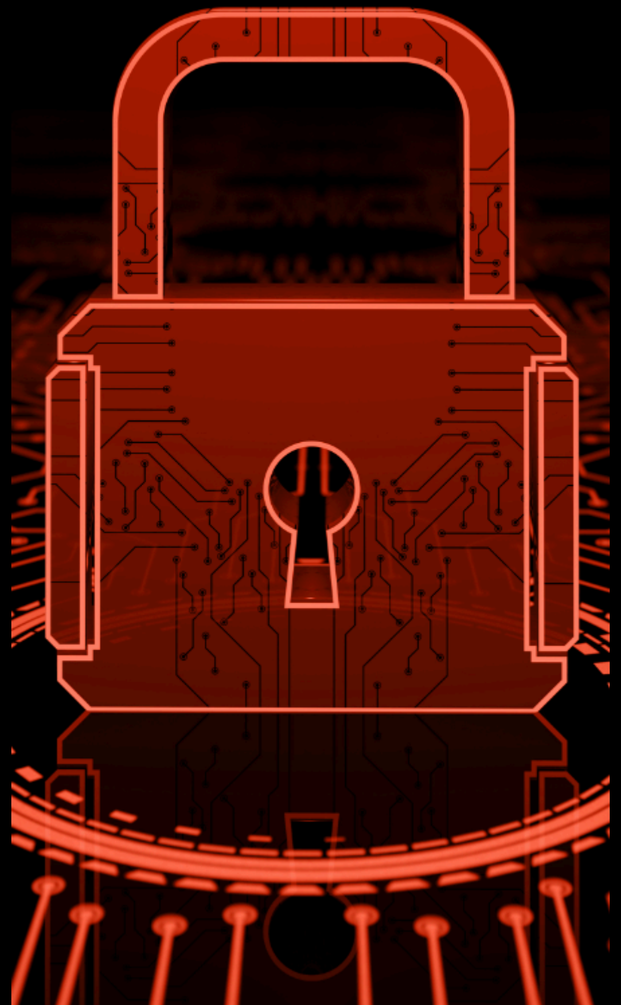
How to Prevent ARP Spoofing :

- Static ARP Entries – Manually mapping IPs to MACs (not scalable).
- ARP Spoofing Detection Tools (e.g., XArp, ArpGuard).
- Network Segmentation – Using VLANs to limit attack scope.
- Encryption (HTTPS, VPN, SSH) – Prevents attackers from reading intercepted data.

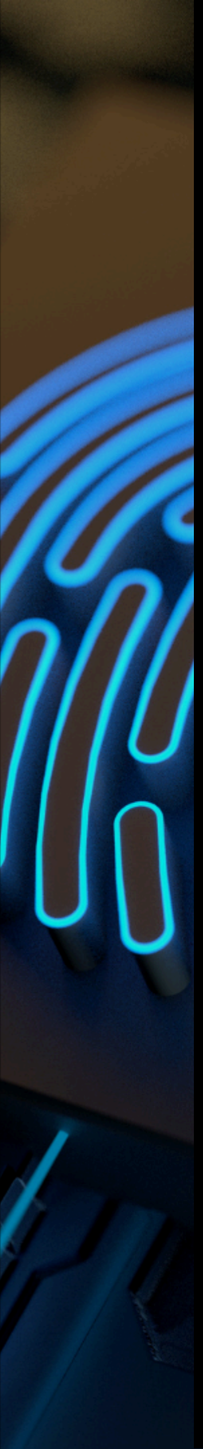
CONCLUSION

ARPSpoofingisa seriousthreat inlocal networks, enabling attackers to redirect, monitor, or manipulate traffic. Prevention involves network monitoring, encryption, and security tools to block unauthorized ARP responses.

In addition to these measures, educating users about safe browsing habits and regularly updating network devices can further fortify defenses against such attacks. It's essential for organizations to stay vigilant and proactive, continuously assessing their network's vulnerability and adapting to new security challenges. By implementing a comprehensive security strategy, businesses can safeguard their valuable data and maintain the trust of their users.







THANK YOU

FOR YOUR ATTENTION